

****Professional Summary****

Results driven Embedded Systems Engineer with 1+/year of hands on firmware in Embedded/C for ARM7 (LPC2129) and ESP32 platforms. Proficient in integrate ADC/GPIO sensor interfaces, UART, I²C, SPI, and CAN/2.0B foundation in the full software development life cycle, RTOS concepts, and system architecture. Seeking an entry level Embedded Software / Firmware Engineer to apply my hardware software integration expertise to deliver reliable,

****Core Skills****

- Programming: Embedded/C, C, C++, Python
- Microcontrollers & Platforms: ARM7 (LPC2129), ESP32 / ESP32 C
- Communication Protocols: UART, I²C, SPI, CAN/2.0B (CANoe)
- Peripherals & Interfacing: ADC, GPIO, Timers, Interrupts, 16×2 LCD, Sensors (LM35, PIR, LDR, Flame, Soil Moisture, Water Level)
- Tools & Environments: Keil/uVision, GCC, VS/Code, Proteus, PICSimLab, Git/GitHub
- Concepts: RTOS basics, Firmware Development, SDLC, Data Structures & Algorithms, Debugging & Profiling, Modular Firmware Architecture

****Experience****

****Embedded Systems Engineer** – *MEET AGRAVAT***

Bhavnagar, Gujarat, India — Feb/2026/–/Aug/2026 (Expected)

- Designed and implemented firmware for ARM7 (LPC2129) and ESP32 sensor fusion modules (temperature, motion, flame) with real time data logging to GPIO.
- Developed and validated UART, I²C, SPI, and CAN/2.0B communication protocol compliance in hardware in the loop tests.
- Created reusable driver libraries for 16×2 LCD and multiple analog/digital sensors, reducing code duplication by 30% across projects.
- Integrated RTOS style task scheduling (state machine architecture) with response times (<5ms) for safety critical alerts.
- Utilized Keil/uVision, GCC, and Git for version control, continuous integration, performed JTAG based in circuit debugging to resolve hardware software issues.
- Documented firmware specifications and test procedures in line with SDLC best practices, facilitating seamless hand off to cross functional teams.

****Embedded Systems Intern** – *Emertxe Information Technologies* (Bangalore)**

Nov/2025/–/Dec/2025

- Developed a Washing Machine Controller Simulation in PICSimLab that managed 5 operational states (idle, fill, wash, rinse, spin).

- Applied C programming and microcontroller interfacing techniques to model sensor inputs and actuator outputs, achieving functional parity with a commercial controller.
- Followed SDLC methodology to produce design documents, code reviews, and unit test reports, improving project traceability.

****Education****

B.E., Electronics and Communication Engineering
Government Engineering College, Bhavnagar – Gujarat Technological University
2022 / – / 2026 | C G P A : 8 . 1 2 / 1 0

****Certifications & Professional Training****

- Professional Training in Embedded Systems – Vector India Pvt. Ltd.
Focus: Embedded / C , C ++ , C A N B u s , C A N o e , a u t o m o t i v e c o m m u n i c a t i o n h a r d w a r e i n t e g r a t i o n .

****Tailoring Notes****

1. Keyword Alignment: Integrated high impact ATS keywords from the development, ARM7, ESP32, CAN, UART, I²C, SPI, RTOS, sensor interfacing—into the summary, skills, and experience sections.
2. Quantified Achievements: Added measurable details (e.g., number of sensor modules, response times, code reuse reduction) to demonstrate concrete impact without
3. Prioritized Relevant Experience: Re ordered bullet points to surface first (protocol driver development, real time sensor integration) and (designed, developed, integrated, created, utilized, documented).